

Competency

- To find the relation between the sets and solve the related problems by demonstrating the relations on the basis of the properties of the sets

Learning Outcomes

- To solve the verbal problems related to the cardinality of sets by using Venn diagrams and solve the behavioural problems by using the properties of relations of sets.

Level-wise learning objectives

S.N.	Levels	Objectives
1.	Knowledge (K)	- To define a set - To tell the types of sets - To state the relations between sets - To list the operations on sets - To define cardinality of sets - To tell the formulae involving two sets
2.	Understanding (U)	- To find the cardinality of a set. - To identify the set notation with special terminologies - To write the word problems based on cardinality relations in set notations
3.	Application (A)	- To solve the verbal problems on operations (union, intersection, difference and complement) of sets by using Venn-diagram - To solve the verbal problems on operations of sets by using formulae
4.	High Ability (HA)	- To relate the problem related to set with other areas of learning like percentage, ratio and so on. - To link various real life/ contemporary problems with sets and solve

Required Teaching Materials/ Resources

Definitions and formulae in colourful chart-paper, scissors, cello tape, different coloured markers, highlighter, models of Venn-diagrams, ICT tools etc.

Pre-knowledge: Check the Pre-knowledge on cardinality of sets, relation of sets, operations of sets and Venn-diagram

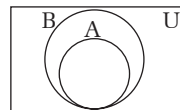
Teaching Activities

1. Make a group discussion on the definition of sets, set notation, types of sets, cardinality of sets and operations of sets by using Venn-diagram.
2. Ask individually the basic concepts from set as revision.

3. Make the group of students and give the questions on set operations from exercise.
4. Solve the problems and verify the relations from the exercise with discussion.
5. Discuss about the cardinality of sets, list the following formula by using Venn-diagrams along with examples. Give the work to the students to write the formulae in chart paper after discussion and paste the best one in math corner of the classroom or math lab.

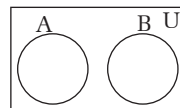
Case-I: When A is a subset of B

- (i) $n(A \cap B) = n(A)$
- (ii) $n(A \cup B) = n(B)$
- (iii) $n_o(B) = n(B) - n(A)$



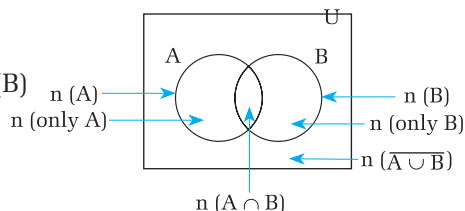
Case- II: When A and B are disjoint sets.

- (i) $n(A \cap B) = 0$
- (ii) $n(A \cup B) = n(A) + n(B)$



Case- III: When A and B overlapping sets

- (i) $n(A \cup B) = n(A) + n(B) - n(A \cap B)$
- (ii) $n(\text{only } A) = n_o(A) = n(A) - n(A \cap B)$
- (iii) $n(\text{only } B) = n_o(B) = n(B) - n(A \cap B)$
- (iv) $n(\text{only one}) \text{ or } n(\text{exactly one}) = n_o(A) + n_o(B)$
- (v) $n(A \cup B) = n_o(A) + n_o(B) + n(A \cap B)$
- (vi) $n(\overline{A \cup B}) = n(U) - n(A \cup B)$
- (vii) $n(U) = n(A) + n(B) - n(A \cap B) + n(\overline{A \cup B})$



6. Explain the useful terminologies in solving verbal problems.

- (i) No. of people who like at least one fruits / either apple or banana = $n(A \cup B)$
- (ii) No. of people who like both apple and banana / who like apple as well as banana = $n(A \cap B)$
- (iii) No. of people who like only apple = $n_o(A)$ and only banana = $n_o(B)$
- (iv) No. of people who like only one fruit = $n_o(A) + n_o(B)$
- (v) No. of people who don't like both the fruits/like neither apple nor banana = $n(\overline{A \cup B})$
- (vi) No. of people who don't like apple only = $n_o(B)$ and no. of people who don't like banana only = $n_o(A)$

Note:

1. If the information are given in percentage, consider that $n(U) = 100$ or x .
2. If the data are given in fraction then suppose that $n(U) = x$.
3. If each people participates in at least one activity then $n(U) = n(A \cup B)$ or $n(\overline{A \cup B}) = 0$

Solution of selected questions from Vedanta Excel in Mathematics

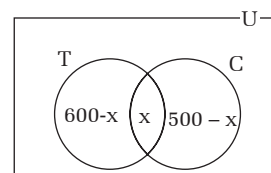
1. *In a survey of 900 students in a school, it was found that 600 students liked tea, 500 liked coffee and 125 did not like both drinks.*
 - (i) Draw a Venn-diagram to illustrate the above information.
 - (ii) Find the number of students who like both drinks
 - (iii) Find the number of students who didn't like tea only.

Solution:

Let T and C denote the sets of students who liked tea and coffee respectively.

Then, $n(U) = 900$, $n(T) = 600$, $n(C) = 500$ and $n(\overline{T \cup C}) = 125$
 Let the no. Of students who liked both the drinks be x .

(i) Drawing a Venn-diagram to show the above information.



(ii) From Venn-diagram,

$$n(U) = (600 - x) + x + (500 - x) + 125 \text{ or, } 900 = 1225 - x = 325$$

Hence, 325 students liked both the drinks.

(iii) No. of students who didn't like tea only i.e., $n_o(C) = 500 - x = 500 - 325 = 175$

2. There are 900 students in a school. They are allowed to cast vote either only for A or for B as their school prefect. 36 of them cast vote for both A and B, 483 cast vote for A and 367 cast vote for B.

(i) How many students did not cast the vote?

(ii) Find the number of valid votes.

(iii) Show the information in Venn-diagram.

Solution:

Here, $n(U) = 900$, $n_o(A) = 483$, $n_o(B) = 367$ and $n(A \cap B) = 36$

Now, we have, $n(A \cup B) = n(U) - n(\overline{A \cup B}) = 900 - 36 = 864$.

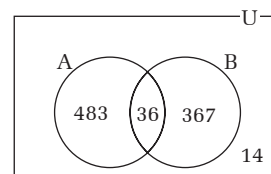
Again, $n(A \cup B) = n_o(A) + n_o(B) + n(A \cap B)$

or, $864 = 483 + 367 + n(A \cap B) \therefore n(A \cap B) = 14$

Hence, 14 students didn't cast the vote.

(i) The number of valid votes
 $= n_o(A) + n_o(B)$
 $= 483 + 367 = 850$

(ii) Illustration in Venn-diagram



3. 54 students of class IX are taking part in sports or in music or in both activities. Out of them 9 students are taking part in both activities. The ratio of the number of the students who are taking part in sports to those who are taking part in music is 5:4.

(i) How many students are taking part in sport?

(ii) How many students are taking part in music only?

(iii) Illustrate this information in a Venn-diagram.

Solution:

Let S and M denote the sets of students who are participating in sports and music respectively.

Then, $n(U) = n(S \cup M) = 54$, $n(S) = 5x$, $n(M) = 4x$ (say) and $n(S \cap M) = 9$

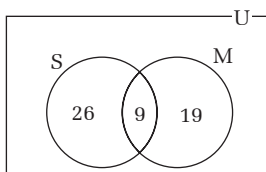
Now, $n(S \cup M) = n(S) + n(M) - n(S \cap M)$ or, $54 = 5x + 4x - 9 \therefore x = 7$

(i) The number of students who are taking part in sport $n(S) = 5x = 35$

(ii) The number of students who are taking part in music $n(M) = 4x = 28$

No. of students who are taking part in sport only, $n_o(S) = n(S) - n(S \cap M) = 28 - 9 = 19$

(iii) Illustration in Venn-diagram



4. In a survey of some students, it was found that 60% of students were studying commerce and 40% were studying science. If 40 students were studying both the subjects and 10% did not study any of two subjects, by drawing a Venn-diagram,
- (i) Find the total number of students.
- (ii) Find the number of students who were studying science only.

Solution:

Let C and S denote the sets of students who were studying commerce and science respectively.

Then $n(U) = 100\%$, $n(C) = 60\%$, $n(S) = 40\%$, $n(\overline{C \cup S}) = 10\%$ and $n(C \cap S) = 40$

Let $n(C \cap S) = x\%$

Now, representing the above data in a Venn-diagram

Again,

From the Venn-diagram,

$$n(U) = n(C) + n(S) - n(C \cap S) + n(\overline{C \cup S})$$

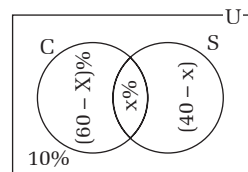
$$\text{or, } 100\% = (60 - x)\% + (40 - x)\% + x\% + 10\% \therefore x\% = 10\%$$

- (i) Let the total number of students $n(U) = x$

$$\text{Then } n(C \cap S) = 10\% \text{ of } x \quad \text{or, } 40 = 0.1x \therefore x = 400$$

Hence, there were 400 students.

- (ii) The number of students who were studying science only $n_o(S) = (40 - 10)\% \text{ of } 400 = 30\% \text{ of } 400 = 120$



Extra Questions

- Out of 30 students of class IX, 15 students like to play volleyball, 20 students like to play basketball and each student like to play at least one of the game.
 - How many students like to play volleyball and basketball both?
 - Show the above information in the Venn-diagram. [Ans: 5]
- Out of 77 districts of Nepal, 27 districts have shared their boarder with India, 15 districts have shared their boarder with China and 37 districts have not shared their boarder with India and China both.
 - How many districts have shared their boarder with China only?
 - How many districts have shared their boarder with India only?
 - Draw a Venn-diagram to show the above information. [Ans: (i) 13, (ii) 25]
- In a group of 30 children, 13 favored apple only, 8 favored guava only and 3 favored none of these fruits, by showing in Venn-diagram
 - Find the ratio of number of children who like both and don't like both the fruits.
 - What percent of the students like guava? [Ans: (i) 2:1, (ii) 60%]
- In a survey of community, it was found that 50% of the people preferred yoga, 60% preferred juggling and 10% preferred neither yoga nor juggling. If 200 people preferred both yoga and juggling, by using a Venn-diagram find:
 - How many people were participated in the survey?
 - How many people preferred only one of these? [Ans: (i) 1000, (ii) 900]